

Phase 2 Insight Report

Analytical Core

Competition: Data Vortex | AARUUSH'26

Theme: Rebuilding the Social Engine

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Date: September 2026

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Executive Summary

This report presents findings from 15 SQL analytical queries executed on the cleaned Social Engine dataset (12,000 posts, 1,500 users, 5 platforms). The analysis covers four dimensions: Trend Detection, Anomaly Discovery, Behavioural Grouping, and Correlation Analysis, supplemented by advanced cross-cutting queries.

1. Trend Detection

1.1 Monthly Post Volume

Post volume remains relatively stable across the 12-month period (May 2024 - April 2025), with counts ranging from 914 (February 2025, lowest) to 1,038 (May 2024, highest).

Month	Post Count	Growth Rate
2024-05	1,038	--
2024-08	1,015	+1.9%
2024-10	1,029	+5.6%
2025-02	914	-9.2%
2025-03	1,013	+10.8%

Insight: The largest drop (-9.2%) occurs in February 2025, followed by the largest rebound (+10.8%) in March. This could indicate seasonal patterns or a data collection gap.

1.2 Weekly Platform Trends

Rolling 4-week averages smooth out noise and reveal that platform-level engagement is remarkably stable. Facebook's rolling average likes fluctuate between ~2,370 and ~2,640, suggesting no single platform experienced a dramatic engagement shift.

1.3 Day-of-Week Patterns

Day	Avg Total Engagement	Rank
Wednesday	3,683	1st
Sunday	3,651	2nd
Saturday	3,642	3rd
Friday	3,584	7th (lowest)

Insight: Mid-week Wednesday leads engagement, while Friday trails. The spread is narrow (99 points between 1st and 7th), indicating that day-of-week is a weak predictor of engagement.

2. Anomaly Discovery

2.1 Viral Post Detection

Using the 2-standard-deviation threshold (mean + 2*stdev), no posts qualified as statistical outliers. This means the engagement distribution is remarkably uniform -- the dataset does not contain extreme viral content. The maximum likes value (5,000) falls within the expected range given the mean (~2,492) and standard deviation (~1,438).

Insight: The uniform distribution suggests the Social Engine's data may be synthetically generated with bounded randomness, or the platform has effective engagement normalization mechanisms.

2.2 Bot Detection

No users exceeded the 3-posts-per-day threshold. With 12,000 posts across 1,500 users over 364 days, the average is ~0.022 posts per user per day, making high-frequency posting extremely unlikely in this dataset.

2.3 Missing Data Analysis

Completeness	Count	Avg Engagement
Complete	8,712	3,620
Platform Missing	1,548	3,642
Text Missing	1,504	3,685
Both Missing	236	3,399

Insight: Posts with only platform or only text missing show *slightly higher* engagement than complete posts, while posts missing both fields show lower engagement. This pattern suggests the corruption was largely random (Missing Completely at Random - MCAR), with the "both missing" group being a small exception.

3. Behavioural Grouping

3.1 User Activity Segments

Segment	Users	Avg Posts	Avg Engagement
Power User	375	11.8	3,697
Active	375	8.7	3,686
Moderate	375	6.9	3,658
Low Activity	375	4.6	3,438

Insight: Power Users produce 2.6x more posts than Low Activity users but only 7.5% higher engagement per post. Volume drives their impact, not per-post quality.

3.2 Platform Loyalty

Dominant Platform	Users	% of Total
YouTube	484	32.3%
Twitter	331	22.1%
Reddit	289	19.3%
Instagram	198	13.2%
Facebook	198	13.2%

Insight: YouTube is the most common "home platform" for users, hosting nearly a third of all dominant-platform users. Instagram and Facebook tie at 13.2% each.

3.3 Cross-Platform Behaviour

Platforms Used	Users	Avg Engagement
1	21 (1.4%)	3,156
2	148 (9.9%)	3,574
3	393 (26.2%)	3,631
4	584 (38.9%)	3,638
5	354 (23.6%)	3,640

Insight: Multi-platform users consistently outperform single-platform users in engagement. The jump from 1 to 2 platforms (+418 avg engagement) is the most significant; returns diminish beyond 3 platforms.

4. Correlation Analysis

4.1 Follower Count vs Engagement

Bracket	Users	Avg Likes
0-5K	131	2,088
5K-10K	156	2,073
10K-20K	320	2,156
20K-30K	299	2,102
30K-40K	301	2,108
40K+	293	2,071

Insight: No meaningful correlation exists between follower count and average likes. All brackets cluster within a 85-point range (2,071-2,156), indicating follower count is not a predictor of post engagement on the Social Engine.

4.2 Brand Sentiment vs Engagement

Sentiment-engagement patterns vary by brand:

- Negative outperforms Positive: Adidas (2,658 vs 2,296), Samsung (2,631 vs 2,480), Pepsi (2,638 vs 2,408)
- Positive outperforms Negative: Coca-Cola (2,563 vs 2,461), Nike (2,508 vs 2,372)
- Near-equal: Apple (~2,500 across all sentiments), Amazon (~2,540 across all)

Insight: Negative sentiment posts often generate higher average likes than positive ones. This "controversy drives engagement" pattern is consistent with real social media dynamics where complaints and critiques attract more attention.

4.3 Location-Language Engagement

Top-performing location-language combinations include Dubai-Arabic (4,497), New York-German (4,492), and Sydney-English (4,468). These outliers involve niche communities within locations, suggesting that language minority groups in certain cities may be more engaged.

5. Advanced Analysis

5.1 Cumulative Engagement

Over 12 months, the Social Engine accumulated:

- 25.4 million total likes
- 12.1 million total shares
- 6.1 million total comments

Monthly accumulation is roughly linear, with no acceleration or deceleration, confirming the platform's stable engagement trajectory.

5.2 Top Users Per Platform

Each platform has distinct top performers, with no user dominating across multiple platforms. Top per-post engagement ranges from 6,375 to 6,966, approximately 1.9x the overall average. These users tend to have moderate follower counts (15K-45K), reinforcing the finding that follower count alone does not predict engagement.

5.3 Account Age

Account maturity shows minimal impact on engagement:

- Accounts < 12 months: 3,602 avg engagement
- Accounts 12-18 months: 3,631 avg engagement
- Accounts 18-24 months: 3,608 avg engagement

Insight: Account age is not a meaningful predictor of engagement quality.

Conclusions

1. Engagement is remarkably uniform across the dataset -- no viral outliers, no bot patterns, minimal variation by day/platform/follower count.
2. Multi-platform presence is the strongest positive correlate with engagement.
3. Negative sentiment often drives higher engagement than positive sentiment for brand-related posts.
4. Missing data appears random (MCAR), suggesting the corruption was not systematic.
5. Follower count and account age are not meaningful predictors of engagement.

SQL Techniques Used

CTEs, Window Functions (LAG, NTILE, ROW_NUMBER, DENSE_RANK, cumulative SUM, RANK), CASE expressions, COALESCE, HAVING, correlated subqueries, JULIANDAY, strftime, statistical calculations (mean, variance, standard deviation, Z-score).